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Two possible ancient dams near the Red Pyramid on the Dahshur Plateau, Egypt

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This study presents, for the first time, evidence of two ancient, giant dams built across Wadi Dahshur, Egypt, along with a water diversion channel west of the Red Pyramid (ca. 4.55–4.53 ka BP). Satellite imagery and watershed mapping show that the two long structures, previously interpreted as stone transport ramps, exhibit the technical signature of dams, each extending 600–700 m and anchored to the valley flanks. The eastern structure likely served as the main water-retention dam, whereas the western structure, equipped with a duckbill weir, may have acted as a check dam, trapping sediment and regulating flow into the diversion channel. Hydrological analyses suggest that Wadi Dahshur connects to western tributaries of Wadi Taflah, potentially granting access to a ~100 km² catchment, possibly ~400 km². The dam's proximity to the Red Pyramid and the nearby diversion canal point to a possible use of water resources during its construction.

During the early IVth Dynasty of Egypt's Old Kingdom (ca. 4.62–4.45 ka BP), monumental pyramid construction entered a new phase under King Snefru^{1,2}, who initiated major building activities at Dahshur with the Bent Pyramid and subsequently erected the Red Pyramid a few hundred meters to the north. The Dahshur plateau was later reoccupied during the Middle Kingdom (ca. 4.02–3.60 ka BP), when at least seven pyramid complexes were built, including the Pyramid of Amenemhet III, establishing the site as a major component of the Memphite funerary landscape^{1,2}.

Beyond monumental architecture, Snefru's reign is also associated with some of the earliest large-scale hydraulic engineering works documented in Egypt. In particular, the Sadd el-Kafara dam^{3–5}, built across the Wadi el-Garawi, about 20 km east of Dahshur, is widely regarded as one of the oldest known large dams in the world and is generally attributed to the early IVth Dynasty, most probably during Snefru's reign³. With an estimated original height exceeding 14 m, a transverse section designed to resist flood loads and faces protected against erosion by cut stone, the dam exemplifies an impressive feat of hydraulic engineering by its builders almost 5000 years ago. Although the Sadd el-Kafara dam ultimately failed during a major flood event³, it provided clear evidence that state-level hydraulic engineering and flood risk management were already underway during Snefru's building program⁶. Other large enclosure walls and hydraulic-related earthworks, such as the Gisir el-Mudir at Saqqara⁷, the Wall of the Crow at Giza⁸, generally interpreted as a flood-control structure designed to protect the workers' settlement from episodic runoff⁹ and the canal and harbor systems documented by Lehner on the Giza Plateau¹⁰, further indicate that water control, flood regulation, and landscape engineering formed an integral part of the broader hydraulic culture of the late IIIrd and IVth Dynasties⁷. Such knowledge and anticipation of environmental processes were essential to

cope with the intense short-term climatic anomalies of the IVth Dynasty, which triggered violent flash floods that repeatedly devastated the site and necessitated urgent reconstruction efforts. The “Lost City” of the Pyramids (Heit el-Ghurab) at Giza—home to the workers and craftsmen who built the Menkaure Pyramid (c. 2500 BCE)—was struck by at least ten major flash floods over a span of approximately 40 to 45 years, each causing severe destruction.

These observations are consistent with the geoarchaeological framework developed by Karl W. Butzer in the 1970s and '80 s, which emphasized the central role of hydrological control, flood management, and landscape engineering in the Nile Valley and its desert margins^{11–13}. Butzer documented the long-term interaction among climatic variability, episodic runoff in desert wadis, sediment dynamics, and early engineering responses, showing that Egyptian hydrological and hydraulic strategies extended beyond the Nile floodplain to include marginal and ephemeral drainage systems when climatic and geomorphic conditions permitted. His work further highlights that early large-scale constructions often served multiple purposes, combining flood mitigation, water storage, and logistical facilitation, rather than strictly agricultural objectives^{11,14}. This perspective is also consistent with broader comparative studies of ancient hydraulic systems, which show that water management, irrigation, flood control, and water-harvesting infrastructures were central components of early state formation, arid-land adaptation, and large-scale construction programs across ancient Egypt^{15–17}.

While the pyramid complexes themselves have been extensively investigated, associated technical infrastructures such as dams¹⁶, flood-control walls, diversion works, canals¹⁵, and workers' installations remain far less well characterized, particularly in terms of their spatial organization,

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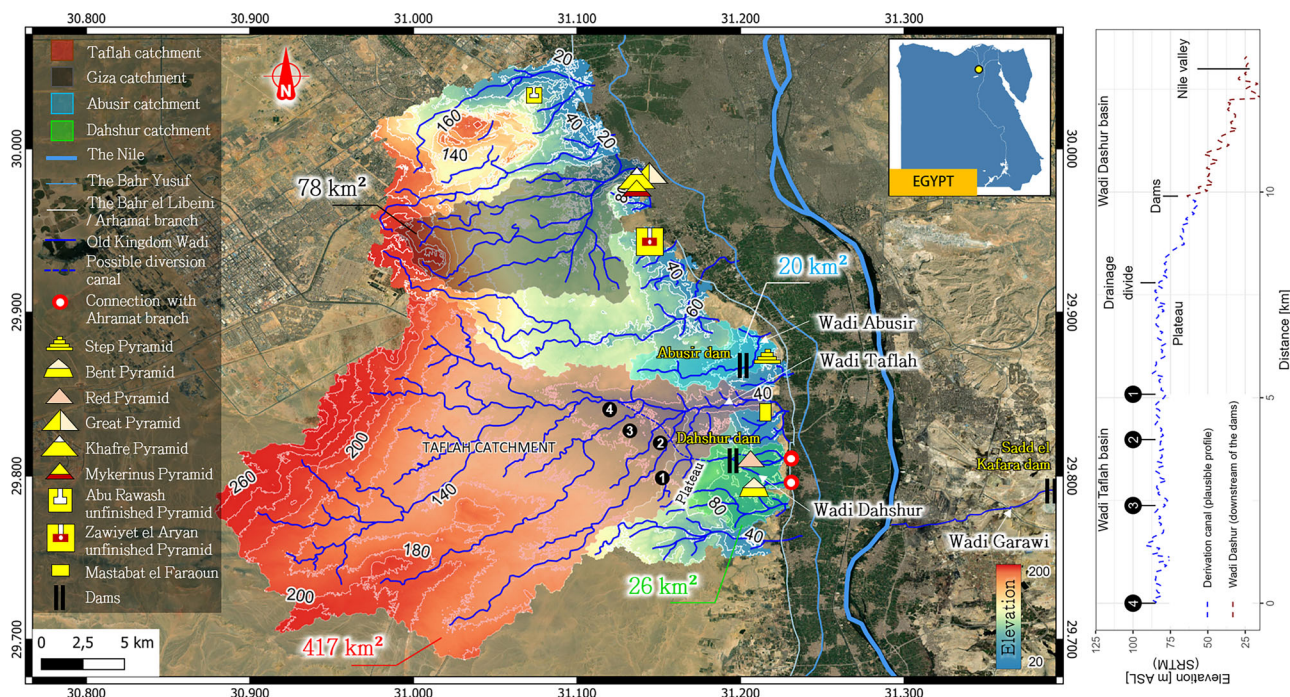


Fig. 1 | Topography and drainage network of the Old Kingdom pyramid plateaus, including major watersheds and sub-watersheds connected to the western Nile plateau. The possible location of the diversion canal is depicted as a blue dotted line

(top panel). The bottom panel shows the longitudinal profiles of the diversion canal and Wadi Dahshur.

geomorphological context, and interaction with surface-water dynamics^{11,18,19}. Their placement within drainage networks, their functional integration at the landscape scale, and their potential role in regulating episodic runoff have rarely been examined in a systematically or comparative manner^{13,20,21}. Recent studies combining paleoclimatic reconstructions and archaeological evidence further indicate that climatic conditions during the early IIIrd and IVth Dynasties (ca. 4.75–4.45 ka BP) were wetter than today and that presently dry desert wadis may have constituted exploitable water resources^{22–24}, suggesting that hydraulic infrastructures could have played a more significant role in construction logistics and site organization than previously assumed.

The Red Pyramid complex is located on the Dahshur plateau (29°49'0" N, 31°13'0" E), on the west bank of the Nile Valley, approximately 30 km south of Cairo as illustrated in Fig. 1. The plateau forms a transitional zone between the Nile floodplain and the western desert and is crossed by the Wadi Dahshur, which provides a natural east–west corridor linking the desert interior to the Nile lowlands. The Bent and Red Pyramids stand at elevations of ~60 m A.S.L. (Above Sea Level), while the wadi floor descends eastward toward the floodplain at ~18–30 m A.S.L. (mean ≈21 m) over a distance of approximately 2 km, corresponding to an average gradient of ~1.5–2%. This moderate but persistent slope is favorable for gravity-driven surface flow and controlled water conveyance. The wadi opens eastward toward Lake Dahshur at the foot of the escarpment, marking the downstream outlet of the drainage system and facilitating lateral transfers of surface water from desert catchments toward the Nile margin. Bebermeier et al.²⁰ describe Dahshur Lake as an oxbow lake, located at the western margin of the Nile floodplain directly at the foot of the escarpment. More recently, Ghoneim et al.²⁵ interpreted Dahshur Lake as most likely the last remaining trace of the abandoned Ahramat Branch of the Nile.

Within this framework, two little-known linear structures crossing the Wadi Dahshur identified on de Morgan's 1897 map²⁶ and shown in Fig. 2), southwest of the Red Pyramid, offer a particularly instructive case study. Their morphology, position within the drainage corridor, and proximity to major IVth Dynasty building activities raise the possibility of a hydraulic

function, which constitutes the central focus of the present investigation. Continuing our previous study⁷, we therefore examine the similarities and differences between the Wadi Dahshur structures and other ancient Egyptian dams by (i) documenting the site, (ii) assessing the plausibility of water availability on the Dahshur plateau during the Old Kingdom and (iii) analyzing the hydraulic functioning of the site. The analysis combines optical and radar satellite imagery, watershed mapping, on-site vegetation records, comparative criteria with ancient dams, and a review of paleoclimatic and hydrological conditions between ca. 5000 and 4500 BP. Although obvious in the landscape, the structures have not yet been excavated. Almost entirely located within a military area, they are difficult to access. Throughout the paper, we highlight elements that would be worth scrutinizing during any eventual excavation to verify, refine, or dismiss our hypotheses and interpretations.

Methods

Study access and interpretive limits

Direct field access to the study area remains constrained by the location of most of the studied structures within a military zone. Consequently, the proposed interpretation relies on geomorphological and remote-sensing analyses, pending ground-truth verification, archaeological excavation, and sedimentological testing.

Digital elevation models and drainage analysis

Two complementary digital elevation models (DEMs) were produced for the geomorphological and hydrological analysis. First, a regional-scale DEM with a resolution of 10 to 30 m was derived from ALOS elevation data and processed using the *Topo to Raster* algorithm after Hutchinson (1989). This DEM was used to analyze the broader topographic setting, regional drainage organization, and the relationship between the Dahshur subcatchments and larger wadi systems. Second, a higher-resolution local DEM was derived from the 1974 topographical map *Le Caire* at a 1:5,000 scale, including sheets H23, H24, H25, H26, H27, F25, F26, G26, and G27, using QGIS 3.24.3. The *Topo to Raster* procedure was selected because it generates hydrologically consistent digital elevation models suitable for drainage and

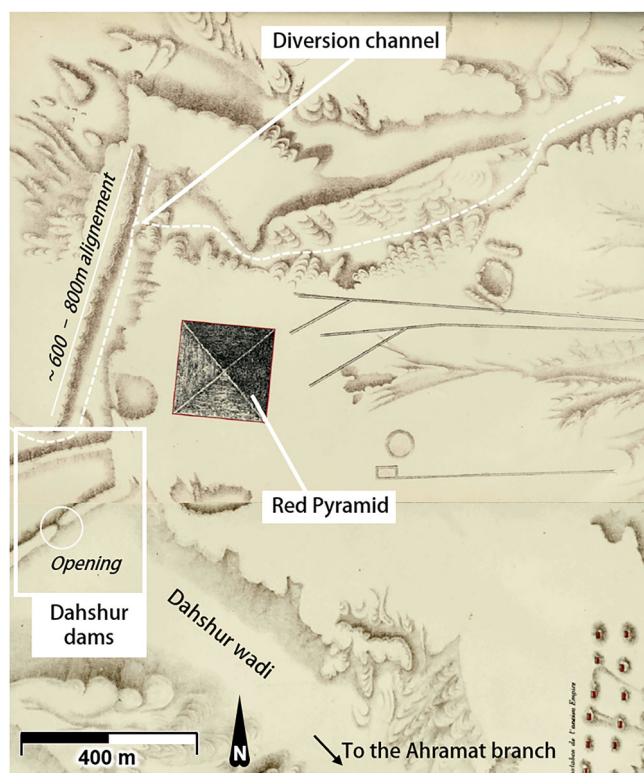


Fig. 2 | Extract from the map “*Carte de la Nécropole Memphite*” drawn by J. De Morgan (1897) shows two dams built across the Wadi Dahshur (bottom left corner), with an opening in the eastern one.

watershed analysis. Derived layers, including slope, aspect, curvature, flow direction, flow accumulation, drainage lines, and catchment boundaries, were calculated in QGIS. The drainage network was digitized and checked against the 1:5,000 topographical maps.

Historical cartography and feature mapping

Historical cartographic sources, including Erbkam (1843), Lepsius²⁷ (1855) and De Morgan²⁶ (1897) maps, were also used to compare formerly mapped features with modern topographic, satellite, and DEM-derived datasets. Linear structures, ancient routes, mapped depressions, drainage alignments, and possible anthropogenic features were georeferenced and compared with present-day surface morphology.

Satellite imagery and paleoclimate context

The analysis further combined optical and radar satellite imagery, watershed mapping, field-based vegetation observations, and a review of paleoclimatic and hydrological conditions between ca. 5000 and 4500 BP, a period that encompasses the Old Kingdom. The paleoclimatic review relied on published regional proxy studies, including evidence for mid- to late-Holocene aridification, Nile floodplain dynamics, Lake Qarun level changes, and broader changes in Saharan and East African hydroclimatic conditions. These data were used to assess the extent to which episodic runoff in local wadis would have been geomorphologically plausible during the period under consideration.

Comparative framework

Finally, a comparative approach was applied to place the two Dahshur linear structures within the broader context of early Egyptian dam and drainage-control systems associated with the IIIrd and IVth Dynasties. The comparison focused in particular on Sadd el-Kafara in Wadi Garawi and the Gisir el-Mudir dam system at Saqqara. This framework was used to assess whether the Dahshur structures are better interpreted

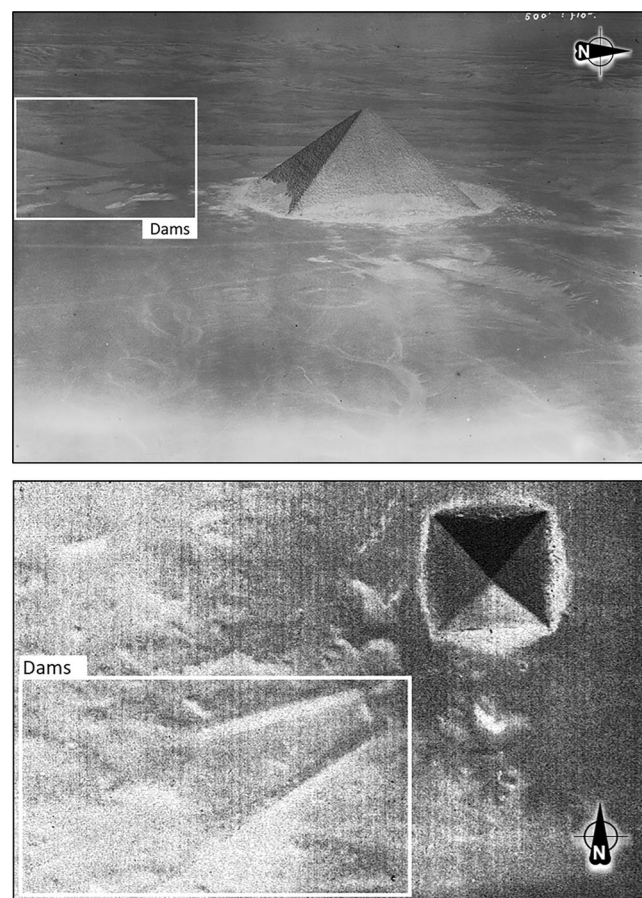


Fig. 3 | Royal Air Force pictures (1925) from the Red Pyramid, with a view of two dams in the background (white rectangle)—coordinates: 29.80667, 31.20650 © UCL Institute of Archaeology. Reproduced with permission.

as natural or transport-related linear features, local drainage-control structures, or check-dam-like installations integrated into Old Kingdom wadi-management practices.

Results

Historical and archaeological reassessment of the Dahshur linear structures and the surrounding constructions

The two linear structures southwest of the Red Pyramid have already been recognized and drawn on the maps of Erbkam (1843), Lepsius²⁷ (1855), and De Morgan²⁶ (1897), as illustrated in Fig. 2. In 1925, the British Royal Air Force photographed these structures from the air (Fig. 3). In 1947, Grinsel interpreted them as two causeways leading from the limestone quarries²⁹ west of the Red Pyramid to its southwestern corner³⁰. In 1981, Arnold also interpreted them as possible transportation roads for the pyramid’s core material³¹ but excluded the possibility that they could have served as ramps, a point debated by Stadelmann^{1,2}. Since their identification, these linear constructions have never been excavated archaeologically and remain buried under a thick layer of aeolian sand. As a result, their precise dimensions are uncertain, although Arnold estimated them to be around 30 m wide and several hundred meters long³¹, a point confirmed by SFS/IGN topographic investigations³², as illustrated in Fig. 4. Since 1975, the site has undergone significant changes, including the construction of a visible pipeline that crosses the area as presented in Fig. 5. In 1997–1998, the army built a fence to the west of the Red Pyramid, and in 2009, an asphalt road was constructed to allow tourist buses and cars to reach the Bent Pyramid. All these constructions were built directly on the two structures. Furthermore, the northern Wadi Dahshur slope had been deliberately flattened, likely during the pipeline’s construction³³.

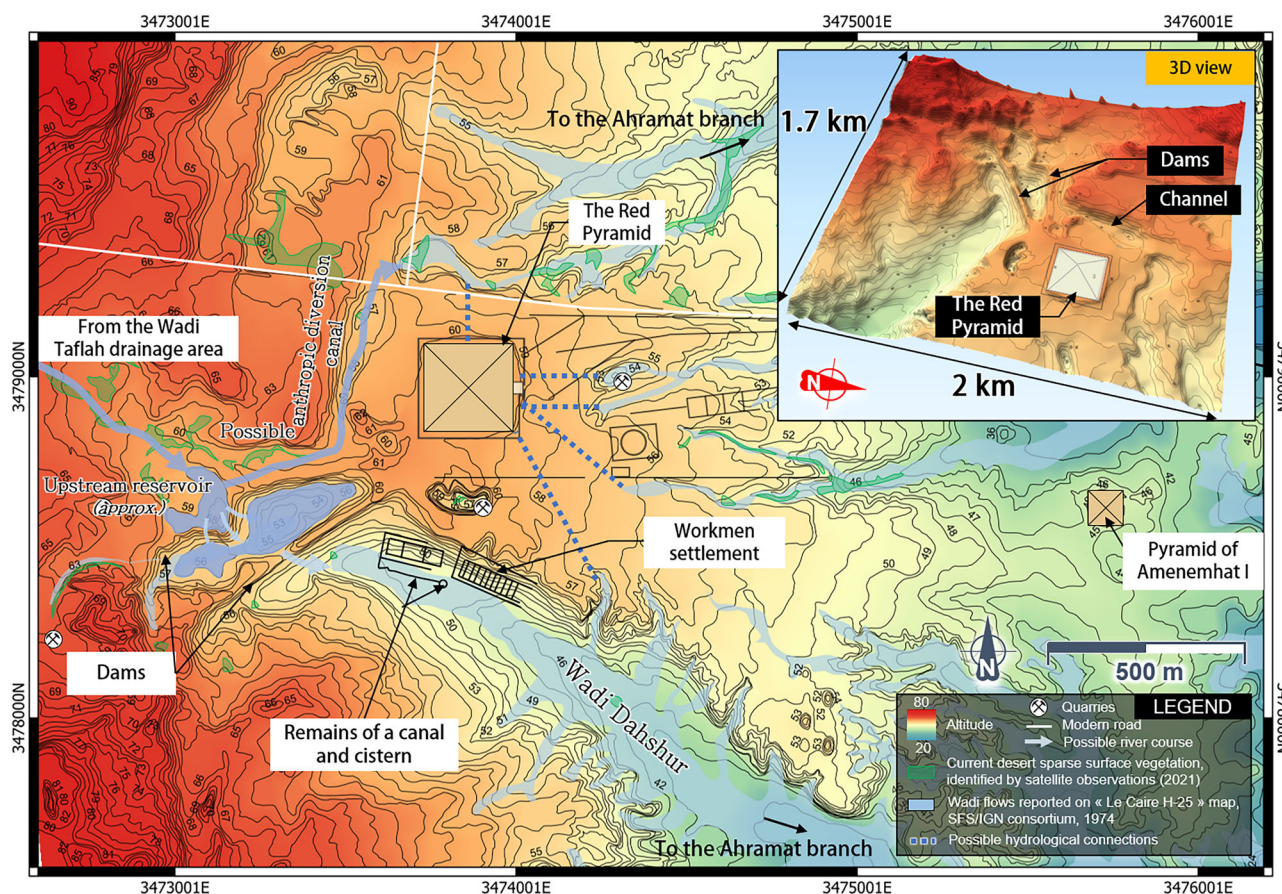


Fig. 4 | Topographical map of the Dahshur plateau. Inset: Digital Elevation Model of the Dahshur dam generated from the 1:5000 topographical map “Le Caire, sheet H25”.

In Bebermeier et al.²⁰, the eastern linear structure is interpreted as a transport ramp linking a quarry area to the pyramid plateau, whereas the western structure is mapped as a natural hill or topographic high. However, de Morgan’s 19th-century map²⁶, reproduced in Fig. 2, clearly depicts an embankment-like feature in the same area. He also describes a ~600–800 m channel to the north, consistent with the scarp lines shown on Bebermeier et al.’s map. As reported by Ramisch et al.¹⁸, the Wadi Dahshur system preserves evidence of ancient anthropogenic reworking, most plausibly related to Old Kingdom construction activities rather than to recent disturbance. Ramisch interpreted anomalies in the relationship between channel-network geometry and surface topography near the pyramid district as indicating channel beds “modified or induced by human impact”, potentially through their use as transport corridors, quarrying, or intentional landscape architecture.

A few dozen meters southeast of the two structures, previous research identified rectangular installations at about 50 m ASL, which Alexanian interpreted as a workmen’s settlement, as shown in Fig. 4^{19,33}. Similar settlements were found along the banks of the Wadi el-Garawi^{34,34}, across which the Sadd el-Kafara dam was built, as shown in Fig. 1. Fireplaces and pottery for provisioning workers indicate that these buildings served as dwellings during the dam’s construction. Apart from the Sadd el-Kafara, Alexanian also found close parallels with workers’ camps and storage buildings near the Heit el-Ghurab wall in Giza, at the Wadi el-Jarf site, or the Wadi el-Araba site³³.

Among other notable findings, Alexanian³³ identified a 165m-long very small canal Fig. 4 in the Wadi Dahshur bed, confirmed by electric resistivity measurements by Kościuk and Ptak^{33,35}. Ernst² also uncovered a possible cistern downstream of the southern structure, as illustrated in Fig. 4, attesting to a water resource. In 2013, similar measurements revealed another canal branch 1500 m downstream in the Wadi Dahshur³⁶, east of

the harbor basin associated with the Bent Pyramid. The wadi’s natural shale ground was found to be 13.80 m ASL, i.e., 12 m below ground level¹⁹. Although previous interpretations favored water input mainly from the Nile backflow during inundation¹⁹, this elevation also implies that the Wadi Dahshur itself could have functioned as an active tributary conveying runoff from its upstream catchment. The channel geometry supports its capacity to temporarily store and transmit water.

North and south of the plateau, the escarpment is dissected by numerous small, west–east ephemeral wadis that drain local subcatchments typically only a few square kilometers in area^{7,37}, as presented in Fig. 1. Here, these subcatchments refer to the small drainage units incised into the escarpment immediately around the pyramid plateau, rather than to the larger regional drainage system. The wadis flowing directly around the Red Pyramid, for instance, drain catchment areas of only about 4–7 km², with drainage divides rising only a few tens of meters above the terrain at the pyramid. These local channels lie immediately east of the much larger Wadi Taflah drainage system, which serves as the principal collector on the west bank of the Nile along the more than 50 km reach where pyramids were built^{38,39}. The Taflah basin extends over approximately 400 km², rises progressively westward to elevations of 200–300 m ASL, and is organized into three main branches as shown in Fig. 1). L-band Synthetic Aperture Radar imagery demonstrates that this network corresponds to an ancient and persistent palaeodrainage system^{37,40} buried beneath aeolian sand cover, still partly visible as paleo channels in the geomorphology and as vegetation alignments along these former channels^{7,41,42}.

Interestingly, both the Saqqara and Dahshur necropolises, where three large pyramids were constructed, are situated in two adjacent valleys in close proximity to the Wadi Taflah drainage system (Fig. 1)). In the case of Saqqara, previous analyses hypothesized the presence of a water diversion from the Taflah network toward the Step Pyramid complex via the Wadi

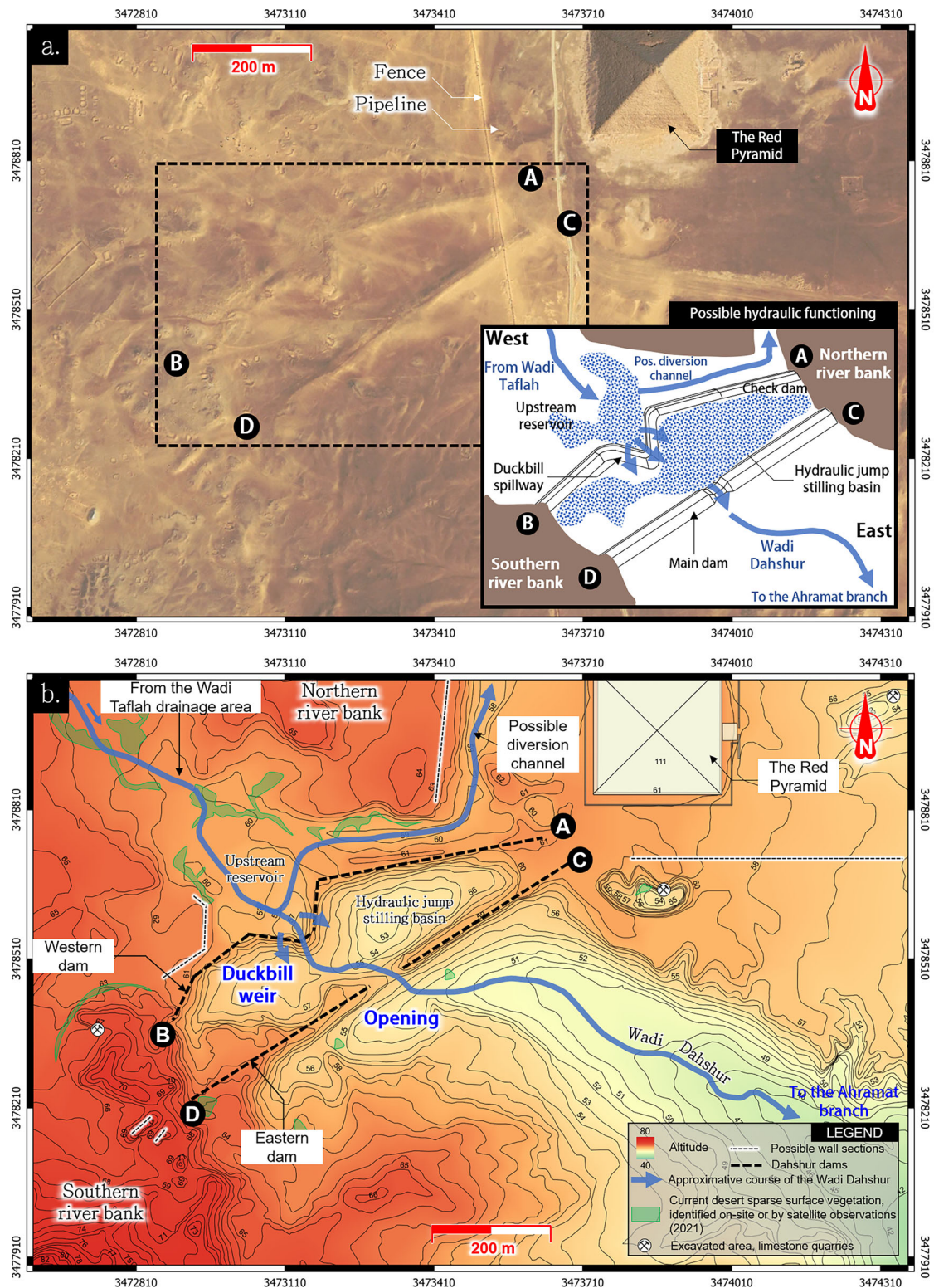


Fig. 5 | The Dahshur dams. a Optical Imagery. **b** Topographical view (Satellite image: Airbus Pléiades, 2021-07-02, reprinted from Airbus D&S SAS library under a CC BY license).

Abusir⁷. We identify a comparable topographic and elevation profile in the area separating the Wadi Taffah from the Wadi Dahshur, as shown in Fig. 1. This drainage divide is singularly flat, located at about 80 m A.S.L., and would be an obvious location for diverting water from the main branches of the Wadi Taffah to a site outside its main valley, thereby protecting it from the flash floods of the whole catchment. As shown in the longitudinal graph

in Fig. 1), points 1 to 4 mark possible intake locations along such a contour-following canal. These points correspond to locations where runoff from branches of Wadi Taffah could potentially have been intercepted and redirected toward Dahshur. Because the canal would have followed a very small gradient (about 1 mm/m is typical for irrigation and small canals worldwide), close to the contour line, multiple intakes could have

progressively increased the effective contributing area supplying Dahshur from only a few square kilometers to over 400 km². In addition to its favorable conditions for quarrying and its proximity to the Nile and to the now abandoned Ahramat and Bahr el-Libeini branches running along the escarpment²⁵, the Dahshur plateau therefore appears strategically positioned to receive runoff from the Taffah branches through short diversion works and simple intake structures. In case of a thunderstorm and flash flood, only a few km² of the upstream catchment, plus a limited amount of water passing through the diversion canal, would have had to be managed at the Dahshur site. The water intakes on the Wadi Taffah could have been very simple weirs, possibly sacrificial structures, requiring repair after major flood events, while the site itself remained in a relatively safe location compared to the Wadi Taffah mouth in the Nile Valley. Such a canal could have been very small, only a couple of meters wide and less than one meter deep, designed to divert discharges ranging from a few dozen to a few hundred liters per second during the wetter winter months. It is therefore not surprising that such a small structure could have disappeared from the landscape after several millennia. Remnants, particularly of water-intake features, may still be preserved along the contour lines just above the drainage divide at 80 m A.S.L. Identifying and documenting such remains will therefore be one of the objectives of future excavations and field investigations.

Revisiting ancient Egypt water resources on the Dahshur plateau

Reconstructing the rainfall regime in the Dahshur region approximately 4700 years ago remains challenging. However, paleoclimatic reconstructions for northern Africa and the eastern Mediterranean^{22,28} indicate that, after the main African Humid Period, precipitation declined from earlier maxima but remained substantially higher than present in many localities, creating conditions conducive to water accumulation. These insights suggest that the Dahshur plateau may have retained exploitable water resources during the Old Kingdom, with potential implications for site hydrology and architectural design.

From 11,000 to 5000 years ago, the Sahara experienced the “Green Sahara” era, when climatic conditions were markedly wetter than today, and landscapes resembled a savannah environment rather than the current desert. This interval supported extensive vegetation, lakes, and fauna adapted to a more humid climate. Paleoenvironmental syntheses by Kuper and Kröpelin²² and by Chandan et al.⁴³ indicate that annual precipitation could exceed ~150 mm across large parts of Egypt, possibly reaching ~250 mm locally.

As this humid phase waned around 5000 BP, precipitation decreased, but not monotonically; instead, climatic conditions remained variable and punctuated by periods of relatively enhanced rainfall²². Such variability is consistent with Holocene reconstructions documenting alternating humid and dry phases across northern Africa and adjacent regions⁴⁴. Speleothem records from Soreq Cave, Judean hills, indicate substantial mid-Holocene rainfall variability, with annual precipitation fluctuating between ~300 and ~700 mm and a major wet peak around 4.8–4.7 ka BP, broadly corresponding to the Old Kingdom²⁴. The Lake Qarun record⁴⁵ in Egypt provides an independent lacustrine archive based on lithofacies, mineralogy, geochemistry, and diatom assemblages, showing high lake levels around ~4.85 ka BP, although this signal partly reflects Nile inflow rather than local rainfall alone.

According to Cilek’s fluvial sediment analyses in the Abusir Lake area 10 km north of Dahshur, the Abusir wadi was probably not merely an episodic drainage feature but a perennial stream during the Old Kingdom⁴⁶. This interpretation is consistent with the broader environmental synthesis proposed by Wong⁴⁷, who emphasizes that Egypt experienced wetter conditions during the Early Dynastic and Old Kingdom periods, with intense and possibly prolonged rainfall episodes affecting several Memphite sites. According to Trzcinski et al.⁴⁸, the strongly cemented L3 structure found in the Great Trench surrounding the Djoser Complex at Saqqara was due to cyclical watering, while the high Fe³⁺ content indicates that the region experienced intensive weathering in a warm, humid environment. In 2020,

Wong further suggested⁴⁷ that the Great Trench, with an estimated capacity of approximately 3.5 Mm³, may have remained permanently water-filled during some periods of the Old Kingdom, pointing to the availability of substantial water resources during this period. Given the ephemeral nature of Mediterranean streams, we doubt that these wadis were perennial at this period. However, this converging evidence suggests that, contrary to the extremely dry conditions that prevailed since this desertification, these wadis experienced occasional flows during the Old Kingdom.

By approximately 4,500 years ago, just before the height of the Old Kingdom, the Sahara and adjacent regions, including Egypt, were entering a phase of intensified aridity. Nevertheless, various paleoclimatic^{22,28,43}, archaeological⁴⁷, and sedimentary studies^{49,50} suggest that parts of Egypt, including areas near the Nile Valley and the Dahshur plateau, still received between 50–150 mm of rainfall annually, essentially in winter. In particular, the synthesis of Chandan et al.⁴³ emphasizes a gradual and spatially heterogeneous decline in rainfall rather than an abrupt shift to hyperaridity, implying that seasonal precipitation and episodic runoff may have remained hydrologically significant in the Nile Valley and its western desert margins during the early Old Kingdom.

Within this climatic framework, Butzer’s geoarchaeological analyses emphasize that societies along the Nile and its desert margins adapted to climatic variability by exploiting episodic runoff and surface water through engineered solutions, rather than relying exclusively on Nile inundation^{11–13}. When combined with isotopic and sedimentary evidence indicating alternating humid and dry phases during the Dynastic period^{24,51} this perspective supports the interpretation that water resources in the Dahshur region may have remained seasonally or intermittently significant despite the long-term trend toward aridity.

Furthermore, although mean annual rainfall could be of the order of 50–150 mm²², rare but intense rainfall events could locally deliver up to 50 mm in a single occurrence, resulting in flash floods. Studies of Egyptian wadis further north, which currently receive similar annual rainfall of approximately 100–200 mm, show that about 1–3% of mean annual precipitation results in surface runoff under normal conditions²⁸. During flash floods, however, runoff coefficients, *i.e.*, runoff volume/rainfall volume, can soar to as high as 30%, generating significant surface water and temporary streams capable of flooding wadis and accumulating in low-lying basins. Although we could not estimate the associated increase in hydrological production, soils in Northern Egypt were likely richer in clay and silt after the Green Sahara period, due to centuries of wetter, savannah-like conditions. These soil characteristics increased the runoff coefficient, meaning that even moderate rainfall could possibly produce substantial surface water flows. As a comparison, the rainfall intensity of the 1994 Egyptian storm at Dronka was 16 mm/h with 22 mm/h for 50 years returns periods and 30 mm/h for 100-years returns periods⁵².

Given such crude rainfall estimates and our limited knowledge of soil types about 5000 years ago, it does not make sense to perform sophisticated hydrological modeling. However, some first-order estimates of water volume and discharge are provided later in the text to give orders of magnitude for how the system could have worked.

Possible hydraulic functioning of the site

Large hydraulic structures are known from Ancient Egypt. In addition to the extensive irrigation and navigation channel network in the Nile valley, the most prominent examples include the Sadd el-Kafara dam^{3–5}, located less than 20 km east of Dahshur, and the Gisir el-Mudir dam, a few 10 km north^{7,53}. These installations were likely intended to regulate sediment and runoff during intense rainfall events, which could otherwise threaten downstream settlements and infrastructure. Beyond flood mitigation, controlled water storage would also have represented a strategic resource for supplying humans and livestock and for supporting large-scale construction activities⁵. In this context, the two linear structures west of the Dahshur plateau display several morphological and geographical characteristics consistent with engineered dam or weir systems. The following sections provide a detailed analysis of these structures.

The Dahshur dams: overview

Digital elevation models (DEMs), historical aerial photographs, and satellite imagery provide additional constraints on the morphology of the two linear structures crossing the Wadi Dahshur (Fig.). Each structure spans the full width of the wadi and is anchored into the opposing valley flanks. Topographic reconstruction indicates lengths of approximately 600–700 m, basal thicknesses of ~15–30 m, and broadly trapezoidal cross-sections, with maximum preserved heights locally reaching ~6–7 m above the current valley floor, levelled by sand filling. They could be higher, but precise dimensions remain uncertain due to deep and extensive aeolian sand cover and restricted access.

A geomorphic analysis of the 1974 topographic maps³² shows that the reservoir upstream of the eastern dam, located in the former wadi valley, was almost filled with sand, with a crest level of about 60 m A.S.L. The area between the two dams, which would form the reservoir for the western dam, has a crest level of 59 m A.S.L., a volume of nearly 230,000 m³, covering about 8.5 hectares, and an average depth of 2.7 m. Since the wadi valley at the toe of this second dam is close to 53 m A.S.L., the reservoir is probably only half filled with sand, meaning its real volume might be twice our initial estimate.

The geometry and spatial arrangement of the two structures are consistent with engineered embankment dams rather than transport ramps or quarry infrastructure. Their transverse orientation across the drainage corridor, lateral anchoring, and trapezoidal profiles correspond to standard stability principles observed in earth and rockfill dams designed to resist hydrostatic loading and erosion⁵⁴, and are comparable to the cross-sectional geometries documented for early Egyptian dam structures such as the Giser el-Mudir⁵³ and the Sadd el-Kafara^{3–5}.

The western structure (Fig. 5a, b, A–B segment) exhibits a surprising zigzagging axis that seems nonsensical on a ramp or road structure. Conversely, such shapes are common on dams and weirs to convey greater discharge for a given cross-wise length. We thus identify this wall as a distinctive central spillway-like geometry approximately 200 m in total crest length, with a zigzagging axis corresponding to a duckbill-type configuration (see later for hydraulic details). The presence of a controlled overflow structure strongly supports a dam function, as regulated spillways are essential to safely evacuate flood discharges and prevent overtopping failure⁵⁵.

The eastern structure (Fig. 5a, b, C–D segment) presents a marked central depression. This feature may correspond to an intentional outlet or auxiliary spillway, a partially eroded breach, or a collapse related to post-depositional processes; this ambiguity can only be resolved through future excavation. From a functional standpoint, such an opening could have facilitated controlled downstream release or periodic sediment flushing, comparable to outlet works in modern detention and sediment-management basins⁵⁶.

The construction of two successive dams within a 200 m reach would be unusual unless they served distinct hydraulic functions. The upstream dam could be a check dam intended to trap sediment, releasing surface water to the second, which would be a water-retention structure⁵⁷. Flow deceleration in the backwater of the first dam would promote the deposition of coarse material and suspended load, thereby limiting downstream sediment transfer to the canals where the intercepted water was sent. In addition, the intermediate basin would act as a dissipation basin, reducing flow energy downstream of the spillway and mitigating scour processes, which are particularly critical in Dahshur marly-limestone substrates^{29,58} that are prone to erosion and undercutting. Multi-stage layouts that combine flow regulation, sediment trapping, and energy dissipation are well implemented in modern hydraulic infrastructure in semi-arid environments and mountainous catchments⁵⁹, and are particularly relevant in Sahelian contexts characterized by episodic high-energy runoff and highly erodible substrates⁵⁷.

Taken together, the two structures may therefore have operated as a coupled hydraulic system, regulating episodic runoff, buffering flood peaks, and managing sediment fluxes before water was delivered to the upstream

diversion channel (see dedicated section) or released in the downstream Wadi Dahshur. Although direct archaeological evidence remains limited, the scale, geometry, and spatial coherence of the installations support their interpretation as large engineered hydraulic works rather than purely logistical transport constructions. Lying upstream of the Bent Pyramid causeway and the associated harbor, they also had an interest in protecting this area from flash floods, as well as the harbor against silting.

The duckbill weir: functional continuity and technical convergence

Analysis of aerial and satellite images reveals that the western dam exhibits a typical triangular duckbill weir structure at its center, as shown in Fig. 5). Spillways with complex shapes, often called labyrinth weirs, composed of several duckbill weirs placed side by side, belong to modern hydraulic engineering developments whereas spillways themselves are much older and are documented in several ancient hydraulic traditions, including Assyrian, Roman, and Cretan systems⁶⁰. In ancient Egypt, however, no securely identified duckbill-type spillway has yet been reported. Consequently, the possible identification of a triangular spillway geometry on a structure tentatively dated to the Old Kingdom period may represent a unique and exceptionally early occurrence of this type of hydraulic configuration. Relevant comparisons include Jawa in Jordan, ca. 3500–3000 BCE⁶¹, Choiromandres in Crete, ca. 1750–1450 BCE⁶², Urartian reservoirs in eastern Anatolia, 8th–7th centuries BCE⁶³, and the Sanchi dams in central India, 3rd–2nd centuries BCE⁶⁴. These examples show that ancient engineers developed diverse solutions for diverting, releasing, or controlling episodic runoff through canals, rock-cut spillways, sluices, and multiple overflow structures. A later but morphologically relevant analog is the Tianbao Weir in Fujian, China, 742–756 CE⁶⁵, whose long-crested, downstream-arching layout has been explicitly compared to modern duckbill or irregular weirs. The apparent absence of comparable features from later Egyptian hydraulic works may reflect the progressive decline of large wadi-management infrastructures following the increasing aridification of Egypt after the IVth Dynasty.

Duckbill weirs are widely employed in contemporary drainage and irrigation systems to maintain stable upstream water levels while safely releasing excess discharge toward downstream channels or outlet/turnout structures⁶⁶, and their hydraulic behavior has been extensively characterized in experimental and applied studies^{67–70}. Compared to a conventional linear weir, the principal advantage of a duckbill configuration lies in its substantially increased discharge capacity per unit channel width under identical water head conditions⁶⁶. This performance gain results from the extended effective crest length provided by the triangular spillway geometry, which allows large flows to be evacuated without excessive upstream water-level rise. In the context of early large dams, this design feature may be interpreted as a technical response to the hydraulic limitations observed in earlier structures. The failure during a strong flood event of the Sadd el-Kafara dam^{3–5}, commonly attributed to absence, undersized or inadequate spillway features during strong flood events, illustrates the risks associated with insufficient flood-release control in early dam engineering. A duckbill-type spillway, by contrast, inherently enhances overflow capacity and reduces the probability of overtopping over unprotected sections of the structure and associated structural instability, suggesting a possible learning process from ancient Egyptians and the progressive refinement of hydraulic design principles.

Weirs are, however, sensitive to sediment accumulation and are typically favored in relatively low-sediment flows. When operated in sediment-laden environments, they commonly require bottom flushing openings or periodic mechanical sediment removal to prevent excessive upstream aggradation. Whether such maintenance strategies existed at Dahshur remains unknown at present and can only be assessed through future archaeological excavation and sedimentological investigation of the structures. Excavating the upstream face of the structure would provide an excellent opportunity to verify whether the deposits are alluvial rather than wind-blown and to obtain absolute dating.

The main dimensional construction parameters for the western dam's duckbill weir can be extracted from the digital elevation model, as illustrated in Fig. 4) and the satellite imagery Fig. 5). Each spillway section has a geometric length of approximately 100 m, giving a total weir length of about 200 m and a height of at least 4 m (up to 7 m). The opening angle is around 70–75°, corresponding to a cycle width of approximately 100 m. This means the ratio of weir crest length to channel width equals two. The discharge capacity of duckbill weirs is obtained from Eq. 1.

$$Q = \frac{2}{3} C_d \sqrt{2g} L_e H^{1.5} \quad (1)$$

Where C_d is the discharge coefficient, L_e is the effective weir length, g is the gravity acceleration, and H is the hydraulic head on the weir crest. C_d depends on the channel and weir geometry, with typical values around 0.5 for a linear weir and 0.62–0.64 for a triangular duckbill weir.

In practical terms, the observed geometry allows the structure to release significantly more water during flood events, potentially up to more than twice as much as a straight weir of the same width, thereby improving flood safety and reducing the risk of overtopping.

The catchment area of Wadi Dahshur at the dam location is 3.2 km², significantly smaller than Wadi Taflah's 416 km² at its outlet (386 km² above the 80 m A.S.L. contour line, where water could be diverted toward the Dahshur dams). A 50 mm rainfall with a 30% runoff coefficient could produce a flash flood of about 50,000 m³ at the dam from Wadi Dahshur, compared with over 6 million m³ at Wadi Taflah's outlet. Without knowing the flood duration, we cannot precisely calculate peak discharge. Assuming a triangular hydrograph lasting an hour, the estimates are about 26 m³/s for Wadi Dahshur and 3500 m³/s for Wadi Taflah. The overflow depth over the dam, calculated by reversing Eq. (1), would be about 0.2 m if only Wadi Dahshur's water is considered, versus roughly 4.5 m if the entire Wadi Taflah flow passes through a similar structure. Proper upstream and downstream protection with cut stone several dozen centimeters thick would allow the dam to withstand the initial flood, but not the larger one. Without these protections, repeated flash floods following the dam's construction would have deposited sediment and led to breaching. During excavation, attention should focus on verifying the types of scour protection upstream and downstream, and on comparisons with dams such as Gisir el-Mudir and Sadd el-Kafara. In essence, even if a small portion of water from Wadi Taflah is diverted during a flash flood, the duckbill weir can handle this discharge without failure.

The diversion channel

The spatial relationship between the duckbill structure and the adjacent diversion canal is shown in Figs. 2, 4 is consistent with conventional hydraulic design principles. Excavation should be carried out to determine if the canal is truly supplied by the western dam or if a small section, currently filled with sand, extends to the intermediate basin. Assuming sufficient water resources, the associated weir would have maintained a relatively stable upstream water level, thereby ensuring a controlled discharge into the channel, while excess water was safely released over the spillway. The channel itself follows a remarkably regular alignment, with at least one straight wall extending over ~600–800 m parallel to the western face of the Red Pyramid before turning sharply eastward, passing within ~40–50 m of the monument. De Morgan's detailed survey²⁶ and later topographic maps further indicate that the canal reconnects with an ancient wadi bed at its eastern termination after a total length of approximately 1 km, as shown in Fig. 2). At its downstream end, this wadi bed is itself connected to the Ahramat Branch²⁵ as illustrated in Fig. 4). The proximity of this hydraulic corridor to two faces of the pyramid, combined with its geometric regularity, identifies this sector as a priority target for future excavation to test the hydraulic interpretation and document construction techniques and sedimentary processes. Over the less than 4 km distance between the dam and the Mastabat el Faraoun, which is located on the bank of the Wadi Taflah mouth (Fig. 1), several additional wadis are situated to the north and east of

the Red Pyramid. These may have served as intermittent drainage channels when the canal overflowed or needed to be drained.

The canal's proximity to the Red Pyramid's western and northern sides naturally raises questions about its possible involvement in the construction process. Assuming sufficient water availability, a controlled canal maintaining a consistent water level would have offered several operational benefits. These include transporting heavy loads, supplying resources, and moving personnel using light barges or water-assisted sleds, along with supporting construction zones and temporary setups. Located a few meters below 60 m A.S.L., this canal stays approximately 40 m above the Nile valley. Ramps were needed to transport supplies from the Nile harbors, but once the canal was filled with water, barges could assist in moving heavy cargo over horizontal distances.

Comparable hydraulic logistics have been documented at Giza, where harbor basins and canals were integrated into pyramid construction systems^{71,72}, and at Wadi el-Jarf, where canalized access and harbor infrastructures supported large-scale state expeditions and material transport^{73,74}. In addition, a controlled water supply could have facilitated ancillary activities such as mortar preparation, dust control, and workforce provisioning. After completion, such water delivery system could be reuse seasonally to feed canal for nearby irrigation or canal for eventual ceremonial use, possibly even feeding the Mastabat el Faraoun area since the terrain from the Red Pyramid to this late IVth dynasty tomb is quite flat with the same altitude of about 60 m A.S.L. While the functional link between the Dahshur canal and pyramid construction remains to be demonstrated archaeologically, its geometry and immediate spatial relationship with the monument justify further targeted investigation.

Discussion

This study introduces, for the first time, a functional interpretation of the two long-recognized linear structures crossing the Wadi Dahshur, considering them as a coupled dam system instead of features related to transport or quarrying. Their spatial organization, transverse positioning across the drainage corridor, lateral anchoring, and associated upstream retention area collectively display the characteristic signatures of engineered embankment dams operating in series. At the landscape scale, watershed analysis and topographic reconstruction Fig. 1) indicates a plausible hydrological transfer between the western tributaries of the Wadi Taflah and the Wadi Dahshur, providing potential access to a contributing catchment of at least ~100 km² and possibly up to ~400 km². Combined with paleoclimatic evidence indicating increased rainfall and episodic runoff during the early Old Kingdom, these findings suggest the presence of a hydrologically meaningful water resource capable of supporting such structures. This resource supplies a few thousand cubic meters annually from the 3 km² natural watershed in Wadi Dahshur, but it could produce several hundred to over one million cubic meters if water is sourced from Wadi Taflah (which receives 50–150 mm/year and has an average runoff coefficient of 2%).

The system's internal setup indicates complementary hydraulic functions. The initial structure may have been a sediment-interception device (currently filled), while the intermediate basin between these structures could have served as a temporary retention and energy-dissipation zone before releasing water downstream in a controlled manner. Such multi-stage arrangements are widely employed in modern flood-control systems in semi-arid environments to mitigate scour, buffer flash floods, and stabilize downstream channels, particularly in erodible lithologies comparable to those observed at Dahshur. Although the precise operational scheme remains to be verified, the overall layout is hydraulically coherent and consistent with engineered flood regulation rather than passive obstruction.

The presence of a high-capacity spillway on the western dam introduces a key functional control. Short-duration storm events in the desert occasionally generate very destructive flash floods. In such regimes, the availability of a dedicated spillway fundamentally changes the failure mode of an embankment, shifting from uncontrolled overtopping and rapid breaching toward controlled overflow and enhanced structural resilience. In this perspective, the adoption of an efficient overflow device may reflect a

progressive empirical refinement of flood-management strategies in response to the hydraulic limitations evidenced by earlier large dams such as Sadd el-Kafara.

Beyond flood regulation, the association of the dam system with a diversion canal running near the Red Pyramid suggests a potential integration of water control within construction logistics and site organization. A regulated canal could have provided stable water levels, reduced sediment delivery, and facilitated waterborne transport, provisioning, or auxiliary technical uses. Comparable canal-harbor systems documented at Giza and other Old Kingdom sites demonstrate that water infrastructure formed an integral component of large construction landscapes, even if direct archaeological confirmation at Dahshur remains pending.

Several uncertainties remain. The absolute chronology of the two dams has not yet been established, and it is not known whether they are strictly contemporaneous with the construction of the Red Pyramid or reflect earlier or later phases of hydraulic activity. Consequently, the availability of water resources during their period of use cannot be assessed with precision. This uncertainty is particularly important because paleoclimatic studies indicate that the IIIrd and IVth Dynasties fall within a period of strong hydroclimatic variability. Chronological offsets of several centuries, which may also concern Old Kingdom monuments in light of recent discoveries⁷⁵, would therefore substantially affect any reconstruction of local water availability. The nature of the central depression observed on the eastern dam—whether an intentional outlet, a structural collapse, or an erosional breach—cannot be resolved without excavation. Likewise, the degree of sediment management, maintenance practices, and long-term operability remains speculative in the absence of stratigraphic and sedimentary data.

Nevertheless, the convergence of geomorphological evidence, watershed connectivity, hydraulic plausibility, and regional cultural context supports the interpretation of the Dahshur structures as intentional large-scale hydraulic installations rather than incidental landscape features. More broadly, these results reinforce the view that water management, including flood regulation and controlled surface-water exploitation beyond the Nile floodplain, may have played a more significant role in Old Kingdom engineering strategies than traditionally assumed. Other hydraulic structures are most probably lost in the desert, waiting to be identified as such.

Future research should prioritize targeted excavation of critical sectors, including the spillway zone, the central sector of the eastern structure, the intermediate basin, and the diversion canal. High-resolution sedimentological, geotechnical, and chronological analyses would allow examination of construction techniques, operational functioning, and temporal attribution, and would help verify good scour anchoring on the valley sides and ensure that the downstream faces are scour-protected down to the valley bedrock to prevent undercutting. At a wider scale, systematic comparison with other early Egyptian hydraulic installations may contribute to a more integrated understanding of the emergence of complex water engineering during the formative phases of Egypt's development.

Data availability

We do not have any research data outside the submitted manuscript file. No custom code was generated or used for this research.

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Author contributions

This paper is based on an original idea of X.L., who wrote most of it. G.P. helped in the hydrological and hydraulic analyses. S.H. reviewed the analysis. All authors approved the submitted manuscript.

Competing interests

Sayed Hemeda is an Associate Editor of npj Heritage Science. He was not involved in the journal's review of, or decisions related to, this manuscript. The authors declare no other competing interests.

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